

AI Executive Assistant Agent

Email management and meeting scheduling through intelligent automation



Sprint Goal Recap

Objective

- Read Gmail emails
- Classify email intent using LLM (Groq)
- Extract informations (date, time, sender)
- Normalize time expressions into structured datetime format.



Technical Decisions

LLM for Understanding Only

- LLM: Semantic understanding
- Python: Deterministic logic

LLM should not calculate time, only extract meaning

Date Strategy

- `dateparser` for robust fallback parsing

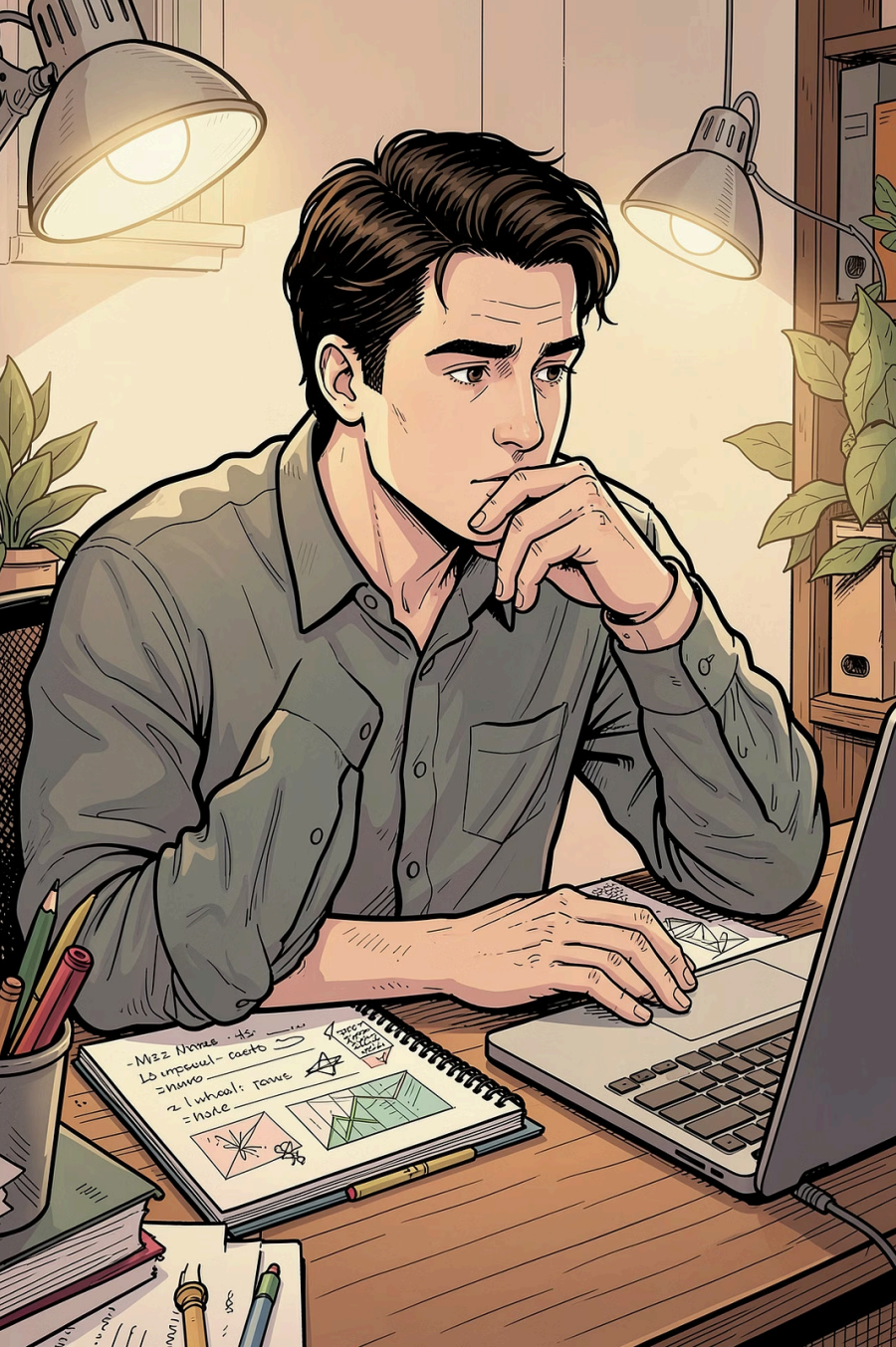
Hybrid System Architecture

A robust hybrid approach combines the strengths of various components for optimal performance and reliability.

- LLM: intent classification & raw datetime extraction
- Python: Date normalization

Gmail Integration

- OAuth authentication for secure access
- Token persistence (saved in `token.json`) to avoid repeated logins
- Automatic handling of read and unread email states



Learnings

What surprised

- LLM does NOT reliably compute dates
- raw text extraction is more stable than structured output

Biggest challenge

- Date normalization ambiguity
- LLM sometimes returns the date the email was received

Key insight

| Hybrid AI systems are more reliable than pure LLM pipelines.

Next Sprint Plan

Planned features for the next sprint:



Improved filtering

ignore newsletters / spam



Google Kalender integration

check calendar availability



Draft responses

Generate email replies and propose suitable time slots



Live demo

Expected output

```
===== AI ANALYSIS =====  
{  
  'intent': 'meeting_request',  
  'confidence': 0.9,  
  'raw_datetime': 'Tomorrow',  
  'sender': 'Farelle Yonzue',  
  'datetime': '2026-05-23 09:00:00'  
}
```

Key moments of the demo

- Email is fetched from Gmail
- LLM extracts intent + raw time
- Python converts "Tomorrow" → real datetime

Open Questions / Feedback Needed

- What can be the best way to handle reliably the phishing mails?
- Should the agent also create events in the calendar?
- UI necessary? (where the user can review suggested time slots, approve draft replies, and manage preferences before anything is sent)